

HDOC - Day 33

Associative Properties of Matrix Addition and Multiplication

Algorithm

1. Display the menu. 2. For addition, read three $m \times n$ VLAs A, B and C; compute $(A+B)+C$ and $A+(B+C)$, display both and compare. 3. For multiplication, read $A(m \times n)$, $B(n \times p)$, $C(p \times q)$; compute $(A \times B) \times C$ and $A \times (B \times C)$. Both final matrices are $m \times q$. 4. Display both sides and compare corresponding elements. 5. State whether the property is verified. 6. Repeat until Exit.

Test Case Table

T C	Operation	Expected	Actual	Result
1	Addition, 2 x 2	LHS = RHS	LHS = RHS	Pass
2	Multiplication, all dimensions 2	LHS = RHS	LHS = RHS	Pass
3	Invalid choice	Error	Error	Pass

C Program

```
#include <stdio.h>
void readM(int r, int c, int a[r][c]){int i, j; for(i=0; i<r; i++) for(j=0; j<c; j++) scanf("%d", &a[i][j]);}
void addM(int r, int c, int a[r][c], int b[r][c], int x[r][c]){int i, j; for(i=0; i<r; i++) for(j=0; j<c; j++) x[i][j]=a[i][j]+b[i][j];}
void mulM(int r1, int c1, int c2, int a[r1][c1], int b[c1][c2], int x[r1][c2]){
    int i, j, k; for(i=0; i<r1; i++) for(j=0; j<c2; j++) {x[i][j]=0; for(k=0; k<c1; k++) x[i][j]+=a[i][k]*b[k][j];}
}
void showM(int r, int c, int a[r][c]){int i, j; for(i=0; i<r; i++) for(j=0; j<c; j++) printf("%d ", a[i][j]); printf("\n");}
int sameM(int r, int c, int a[r][c], int b[r][c]){int i, j; for(i=0; i<r; i++) for(j=0; j<c; j++) if(a[i][j]!=b[i][j]) return 0; return 1;}

void addAssoc(void){
    int m, n; printf("Enter m n: "); scanf("%d%d", &m, &n);
    int A[m][n], B[m][n], C[m][n], AB[m][n], BC[m][n], L[m][n], R[m][n];
    printf("A: "); readM(m, n, A); printf("B: "); readM(m, n, B); printf("C: "); readM(m, n, C);
    addM(m, n, A, B, AB); addM(m, n, AB, C, L); addM(m, n, B, C, BC); addM(m, n, A, BC, R);
    printf("LHS (A+B)+C: \n"); showM(m, n, L); printf("RHS A+(B+C): \n"); showM(m, n, R);
    printf(sameM(m, n, L, R)?"Associative property of addition is verified.\n":"Not verified.\n");
}

void mulAssoc(void){
    int m, n, p, q; printf("Enter m n p q: "); scanf("%d%d%d%d", &m, &n, &p, &q);
    int A[m][n], B[n][p], C[p][q], AB[m][p], BC[n][q], L[m][q], R[m][q];
    printf("A: "); readM(m, n, A); printf("B: "); readM(n, p, B); printf("C: "); readM(p, q, C);
    mulM(m, n, p, A, B, AB); mulM(m, p, q, AB, C, L); mulM(n, p, q, B, C, BC); mulM(m, n, q, A, BC, R);
    printf("LHS (A*B)*C: \n"); showM(m, q, L); printf("RHS A*(B*C): \n"); showM(m, q, R);
    printf(sameM(m, q, L, R)?"Associative property of multiplication is verified.\n":"Not verified.\n");
}

int main(){int ch; do{printf("\n1. Associative addition\n2. Associative multiplication\n3. Exit\nEnter choice: "); scanf("%d", &ch);
    switch(ch){case 1: addAssoc(); break; case 2: mulAssoc(); break; case 3: break; default: printf("Invalid choice.\n");} while(ch!=3); return 0;}
```

Compilation, Execution & Testing

```
puppie@Anvikas-MacBook-Air Desktop % clang day33.c -o day33
puppie@Anvikas-MacBook-Air Desktop % ./day33
Enter choice: 1
Enter m n: 2 2
A: 1 2 3 4 B: 5 6 7 8 C: 1 1 1 1
LHS (A+B)+C: 7 9 / 11 13
RHS A+(B+C): 7 9 / 11 13
Associative property of addition is verified.
```

```
puppie@Anvikas-MacBook-Air Desktop % ./day33
Enter choice: 2
Enter m n p q: 2 2 2 2
A: 1 2 3 4 B: 1 0 0 1 C: 5 6 7 8
LHS (A*B)*C: 19 22 / 43 50
RHS A*(B*C): 19 22 / 43 50
Associative property of multiplication is verified.
```

Result: The associative properties of matrix addition and multiplication were successfully verified by comparing the matrices obtained on both sides.